Mini Project II

Serverless & Container Management

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Laporan ini dibuat untuk memenuhi nilai tugas *Mini Project* II di *Bootcamp* kelas *Cloud Engineer: From Fundamental to Deployment.*

Soal *Mini Project* 15 April 2026 :

1. *Serverless Function.*

Buat aplikasi sederhana menggunakan *AWS Lambda* atau *Google Cloud Functions.* Fungsi cukup melakukan operasi dasar seperti *Create, Read,* atau *Hello World* dengan respons dalam format *JSON.*

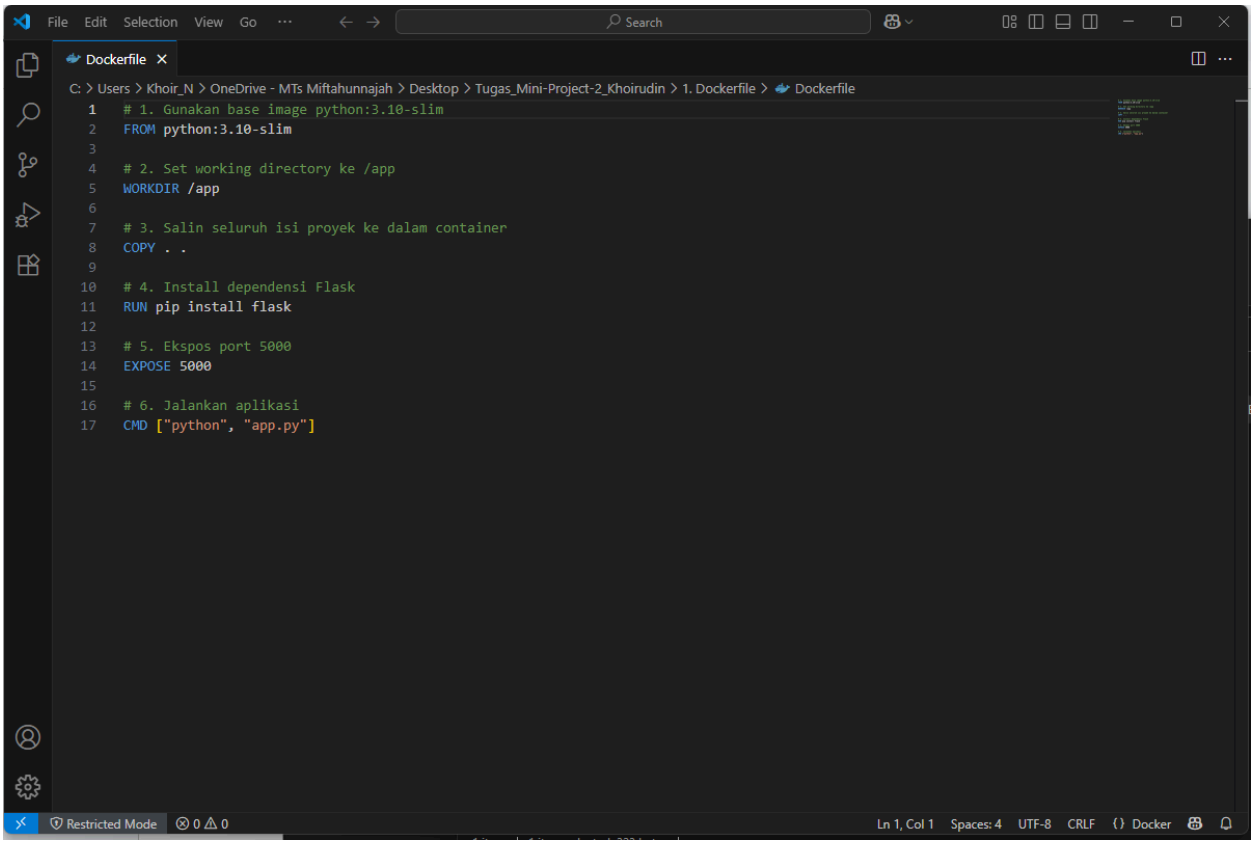
2. *Container Management* dengan *Docker.*

Kamu diberikan *source code* aplikasi *web ToDo* berbasis *Python* dan *Flask*. Aplikasi ini menyimpan data dalam *file* lokal (todos.json) dan tidak menggunakan *database*.
Source Code:
https://github.com/maliqpotter/cloud-engineer-digital-skola/tree/sesi-11

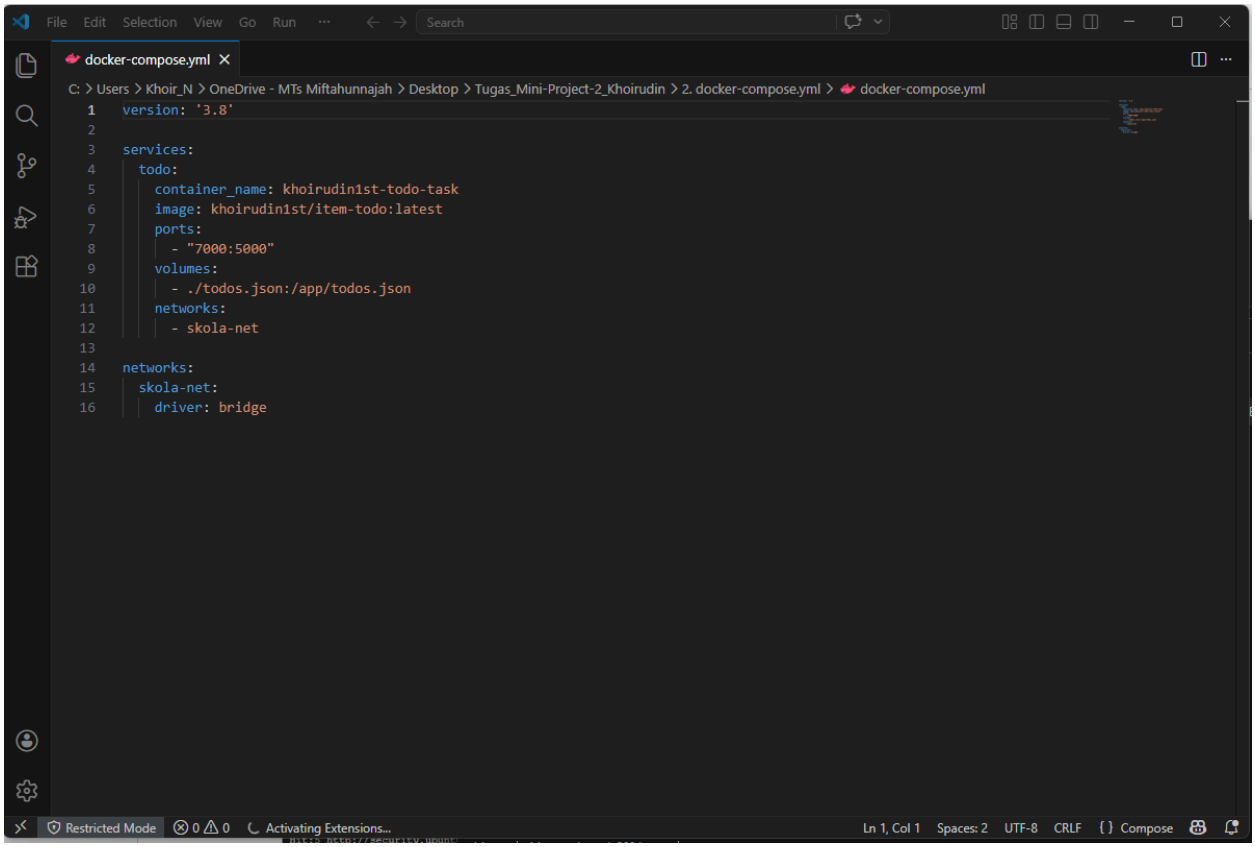
Tugas kamu meliputi:
1. *Dockerisasi* aplikasi menggunakan *Dockerfile*.
2. *Push image* ke *container registry* (misalnya *Docker Hub*).
3. Susun file *docker-compose.yml* untuk menjalankan aplikasi menggunakan *image* tersebut.
4. Akses aplikasi melalui *port* 7000 di *host*.
5. Gunakan jaringan *custom* bernama skola-net.

Berkas yang Harus Dikumpulkan :

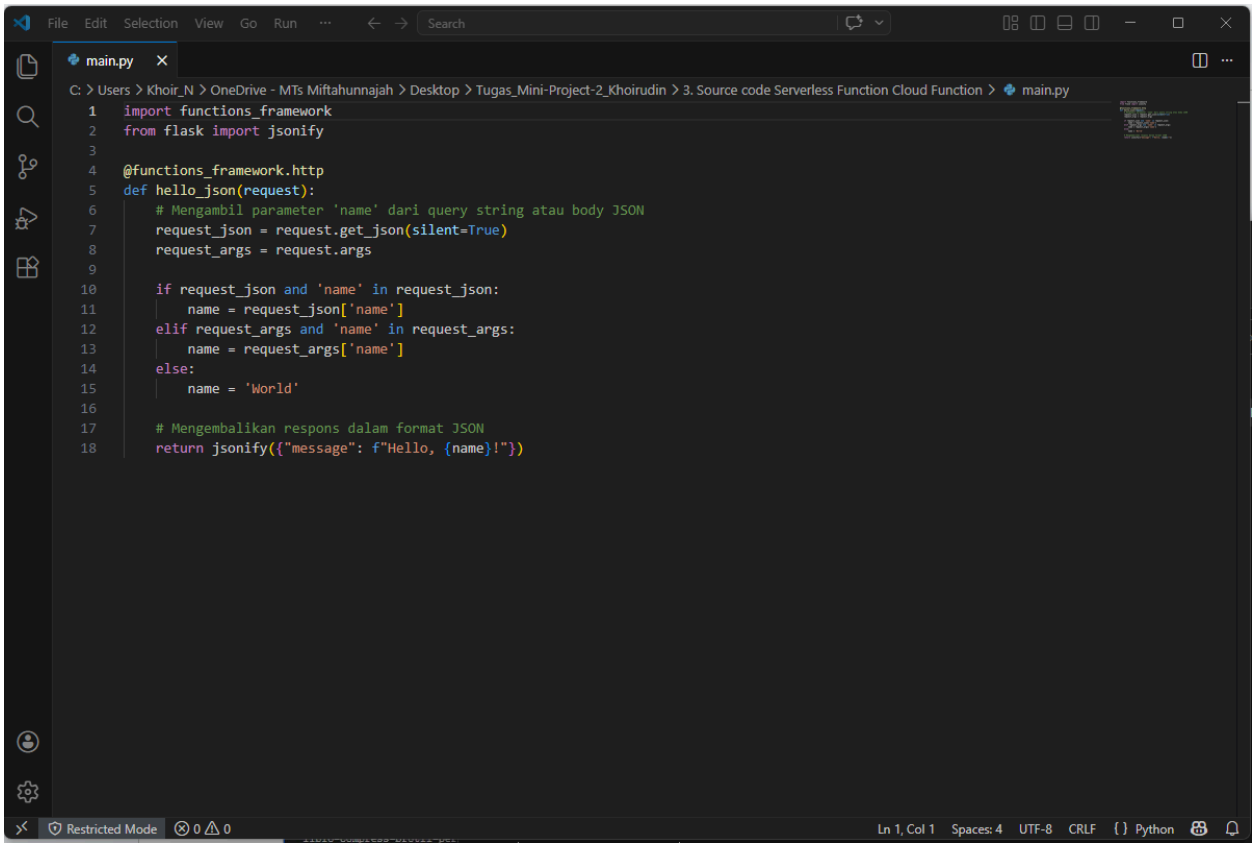
1. Dockerfile.



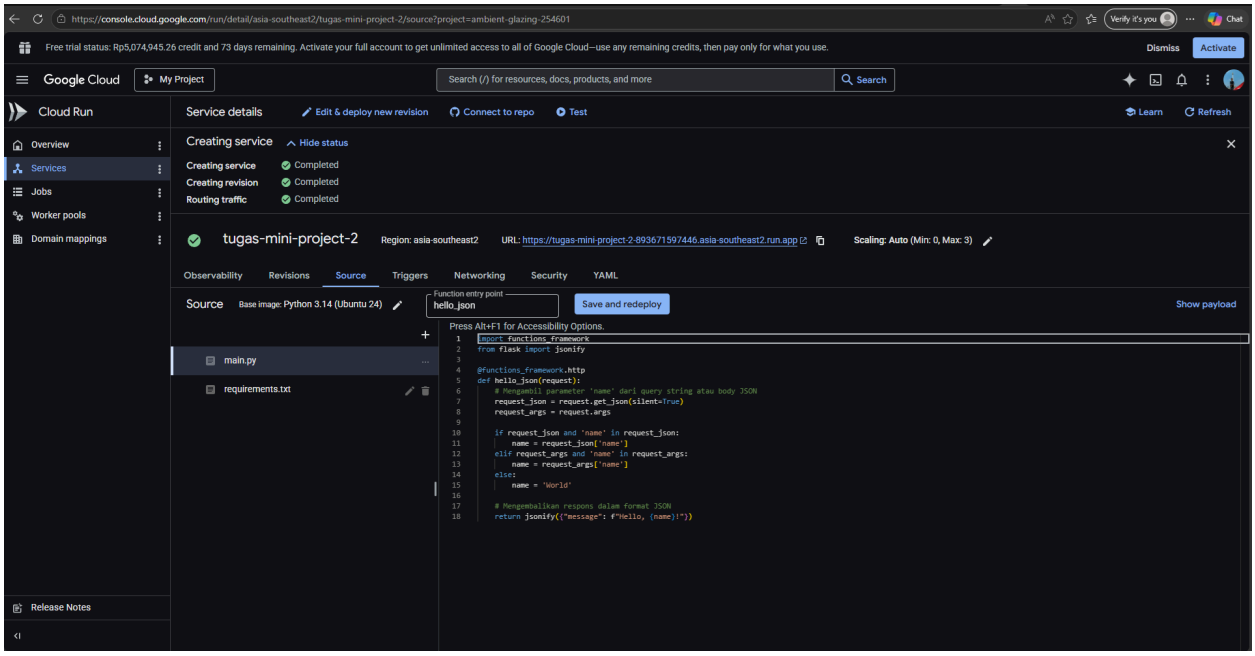
2. docker-compose.yml.

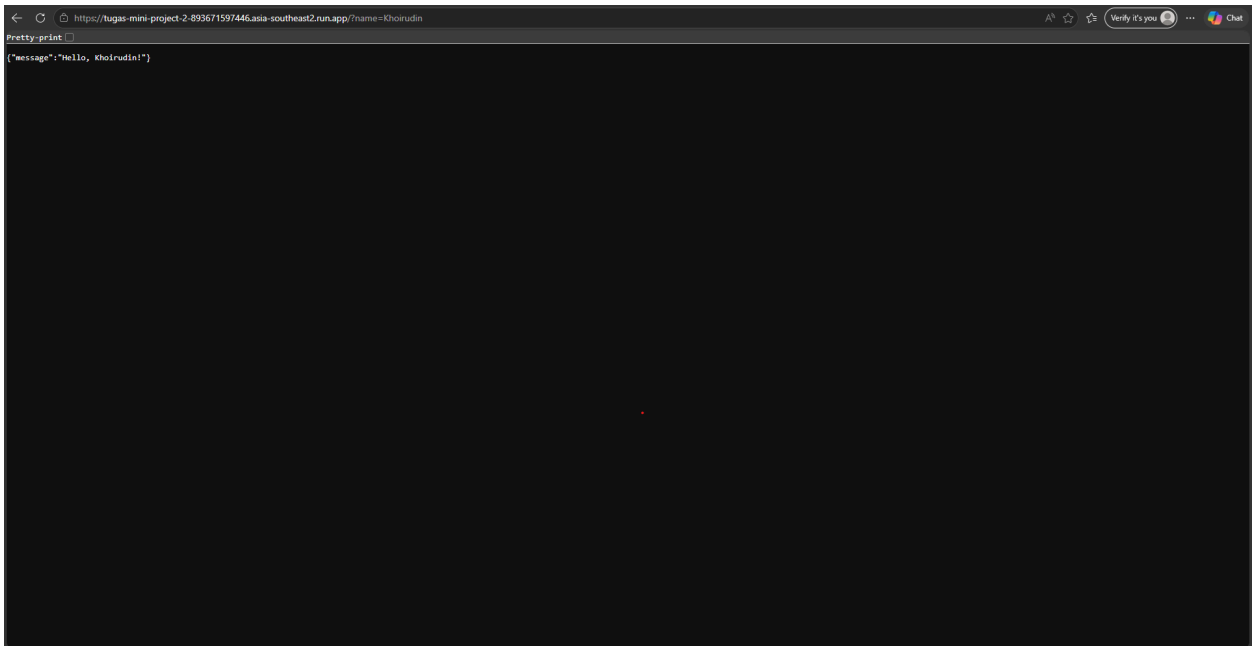


3. Source code Serverless Function (file utama Lambda / Cloud Function).



4. Screenshot.
    a. Fungsi Serverless berjalan dan URL endpoint.





b. Build image Docker.

```
Windows PowerShell

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\Khoir_N\OneDrive - MTs Miftahunnajah\Desktop\Tugas 2\tugas-docker-skola> docker build -t khoirudin1st/item-t
odo:latest
ERROR: docker: 'docker buildx build' requires 1 argument

Usage:  docker buildx build [OPTIONS] PATH | URL | -

Run 'docker buildx build --help' for more information

What's next:
    Debug this build failure with Gordon → docker ai "help me fix this build failure"
PS C:\Users\Khoir_N\OneDrive - MTs Miftahunnajah\Desktop\Tugas 2\tugas-docker-skola> docker build -t khoirudin1st/item-t
odo:latest .
[+] Building 10.8s (5/9)                                                                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                0.1s
 => => transferring dockerfile: 362B                                                                                0.0s
 => [internal] load metadata for docker.io/library/python:3.10-slim                                                 3.2s
 => [auth] library/python:pull token for registry-1.docker.io                                                       0.0s
 => [internal] load .dockerignore                                                                                   0.0s
 => => transferring context: 2B                                                                                     0.0s
 => [1/4] FROM docker.io/library/python:3.10-slim@sha256:cdbf8193cee2e31639ea8ea85ffdd8fa5cce98ee9abfde96ea5f3294  7.4s
 => => resolve docker.io/library/python:3.10-slim@sha256:cdbf8193cee2e31639ea8ea85ffdd8fa5cce98ee9abfde96ea5f3294  0.0s
 => => sha256:6f7b1a1af1d212ff1519dbf4534d2491d175f4fa052f008e4cfc2ad9744e5e4b 249B / 249B                         0.6s
 => => sha256:b05032fa5b62ec785bb4d899754c0969d9829fa10bbdc7f74bb14c8e4f11358a 13.82MB / 13.82MB                   6.9s
 => => sha256:b4e72df0ba0e7a032d9028786a436c63d283a51c04415ae0168197be7d828938 1.29MB / 1.29MB                     1.7s
 => => sha256:3531af2bc2a9c8883754652783cf96207d53189db279c9637b7157d034de7ecd 16.78MB / 29.78MB                   7.2s
 => [internal] load build context                                                                                   0.1s
 => => transferring context: 2.50kB                                                                                 0.0s
```

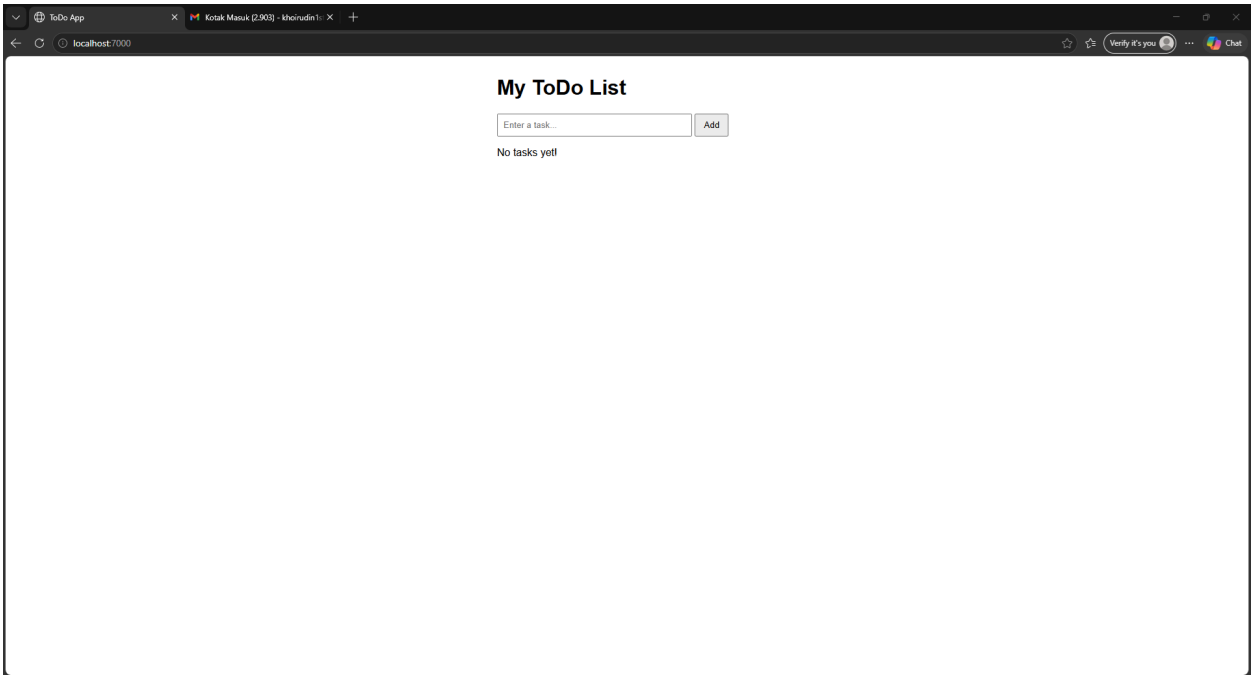
```
Windows PowerShell

 => [internal] load build definition from Dockerfile                                                                0.1s
 => => transferring dockerfile: 362B                                                                                0.0s
 => [internal] load metadata for docker.io/library/python:3.10-slim                                                 3.2s
 => [auth] library/python:pull token for registry-1.docker.io                                                       0.0s
 => [internal] load .dockerignore                                                                                   0.0s
 => => transferring context: 2B                                                                                     0.0s
 => [1/4] FROM docker.io/library/python:3.10-slim@sha256:cdbf8193cee2e31639ea8ea85ffdd8fa5cce98ee9abfde96ea5f329  12.6s
 => => resolve docker.io/library/python:3.10-slim@sha256:cdbf8193cee2e31639ea8ea85ffdd8fa5cce98ee9abfde96ea5f3294  0.0s
 => => sha256:6f7b1a1af1d212ff1519dbf4534d2491d175f4fa052f008e4cfc2ad9744e5e4b 249B / 249B                         0.6s
 => => sha256:b05032fa5b62ec785bb4d899754c0969d9829fa10bbdc7f74bb14c8e4f11358a 13.82MB / 13.82MB                   6.9s
 => => sha256:b4e72df0ba0e7a032d9028786a436c63d283a51c04415ae0168197be7d828938 1.29MB / 1.29MB                     1.7s
 => => sha256:3531af2bc2a9c8883754652783cf96207d53189db279c9637b7157d034de7ecd 29.78MB / 29.78MB                  10.8s
 => => extracting sha256:3531af2bc2a9c8883754652783cf96207d53189db279c9637b7157d034de7ecd                           0.8s
 => => extracting sha256:b4e72df0ba0e7a032d9028786a436c63d283a51c04415ae0168197be7d828938                           0.1s
 => => extracting sha256:b05032fa5b62ec785bb4d899754c0969d9829fa10bbdc7f74bb14c8e4f11358a                           0.6s
 => => extracting sha256:6f7b1a1af1d212ff1519dbf4534d2491d175f4fa052f008e4cfc2ad9744e5e4b                           0.0s
 => [internal] load build context                                                                                   0.1s
 => => transferring context: 2.50kB                                                                                 0.0s
 => [2/4] WORKDIR /app                                                                                              0.2s
 => [3/4] COPY . .                                                                                                  0.1s
 => [4/4] RUN pip install flask                                                                                     3.1s
 => exporting to image                                                                                              1.1s
 => => exporting layers                                                                                             0.6s
 => => exporting manifest sha256:ced3de241f85271d91351ffc27cbcbd437110075044544ab0351a95f0fccf84d                   0.0s
 => => exporting config sha256:7f0e488a901cbbb6a4d13ba76b89b9da04b0cf904359e33c9e5fc2cc12368c4e                     0.0s
 => => exporting attestation manifest sha256:c3960898f43c92c43a5c0beb1bc55c0b07a4f848763e80e348119364ad35feb1       0.0s
 => => exporting manifest list sha256:8b0fad2d5a4d7219e99cb05550360aceb5a651267018afb0a18a7481e25519ed              0.0s
 => => naming to docker.io/khoirudin1st/item-todo:latest                                                            0.0s
 => => unpacking to docker.io/khoirudin1st/item-todo:latest                                                         0.4s
PS C:\Users\Khoir_N\OneDrive - MTs Miftahunnajah\Desktop\Tugas 2\tugas-docker-skola>
```

c. Hasil docker ps (list container yang berjalan).

```
Windows PowerShell
241fa574064f: Pushed
23e7426cf359: Pushed
78c06afc0874: Pushed
07a9a86ee84c: Pushed
6f7b1a1af1d2: Pushed
3531af2bc2a9: Pushed
b05032fa5b62: Pushed
b4e72df0ba0e: Pushed
latest: digest: sha256:8b0fad2d5a4d7219e99cb05550360aceb5a651267018afb0a18a7481e25519ed size: 856
PS C:\Users\Khoir_N\OneDrive - MTs Miftahunnajah\Desktop\Tugas 2\tugas-docker-skola> docker-compose up -d
time="2026-05-01T20:54:41+07:00" level=warning msg="C:\\Users\\Khoir_N\\OneDrive - MTs Miftahunnajah\\Desktop\\Tugas 2\\
tugas-docker-skola\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avo
id potential confusion"
[+] up 2/2
 ✔Network... Created                                                                                                0.1s
 ✘ Contain... Error response from daemon: Invalid container name (khoirudin1st/todo-task), only [a-zA-Z0-9][a-zA-Z0-9_.-
] are allowed 0.0s
Error response from daemon: Invalid container name (khoirudin1st/todo-task), only [a-zA-Z0-9][a-zA-Z0-9_.-] are allowed
PS C:\Users\Khoir_N\OneDrive - MTs Miftahunnajah\Desktop\Tugas 2\tugas-docker-skola> docker-compose up -d
time="2026-05-01T21:03:04+07:00" level=warning msg="C:\\Users\\Khoir_N\\OneDrive - MTs Miftahunnajah\\Desktop\\Tugas 2\\
tugas-docker-skola\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avo
id potential confusion"
[+] up 1/1
 ✔Container khoirudin1st-todo-task Started                                                                          0.6s
PS C:\Users\Khoir_N\OneDrive - MTs Miftahunnajah\Desktop\Tugas 2\tugas-docker-skola> docker ps
CONTAINER ID   IMAGE                          COMMAND           CREATED              STATUS              PORTS
                                   NAMES
518aee19dcef   khoirudin1st/item-todo:latest   "python app.py"   About a minute ago   Up About a minute   0.0.0.0:7000->
5000/tcp, [::]:7000->5000/tcp   khoirudin1st-todo-task
PS C:\Users\Khoir_N\OneDrive - MTs Miftahunnajah\Desktop\Tugas 2\tugas-docker-skola>
```

d. Tampilan web aplikasi ToDo yang sudah bisa diakses melalui



Penutup

"Teknologi terbaik adalah yang tak terlihat; ia bekerja dalam sunyi, menghapus jarak antara imajinasi dan realitas."

— ***Anonim***